

Neural Survival Models: LongitudinalDeepHit

Part E(iii) — Transformer Encoder for Competing Risks

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Why go beyond Cox? Mortgage prepayment is path-dependent.

Cox / Andersen-Gill limitations

- Proportional hazards: covariate effect is **constant over time**
- Single scalar score — cannot model the **S-curve shape** of prepayment
- Competing risks require separate model fits; no shared representation
- Cannot exploit the **sequence structure** of monthly covariate history

What a sequence model can do

- Read the full monthly path $[x_1, \dots, x_T]$ — **burnout is encoded**
- Produce nonlinear, time-varying hazard rates $\lambda_k(t | x)$
- Model prepayment and default **jointly** via shared encoder
- Attention weights identify **which months** drive the prediction

A 2007-vintage subprime loan and a 2020-vintage prime loan may have identical origination snapshots but **opposite** trajectory histories — only a sequence model distinguishes them.

Three design decisions: why this architecture over the alternatives.

Alternatives considered

LSTM	Vanishing gradients at $T=120$; attends only to adjacent past month; harder to interpret
Single-context head	Compresses full history to one vector; head has no time index, produces near-constant hazard curve; gradient reaches only the last hidden state
Two separate models	Violates independent censoring: defaulted loans are the riskiest borrowers, not random exits; loses shared macro signal

Why this combination

Causal Transformer	Reaches any past month in one attention step; gradient flows through every position; attention weights interpretable
Per-timestep head	$\lambda(t) = \text{head}(h_t)$: hazard conditioned on full causal history $x_1 \dots x_t$; each month independently trained
Joint encoder	Shared representation learns that same macro shock raises default and lowers prepayment; chain product enforces $\text{CIF}_1 + \text{CIF}_2 + S = 1$

LongitudinalDeepHit: Transformer encoder with two CIF output heads.

Architecture

Input	$(B, T_{\max}=120, D=13)$ padded sequences
Proj.	Linear $D \rightarrow d_{\text{model}}=64$
PE	Sinusoidal positional encoding on loan age
Enc.	$\times 2$ pre-norm layers, 4 heads, causal mask
Head 1	FC $64 \rightarrow 64 \rightarrow 1$ per h_t , Sigmoid $\Rightarrow \lambda_1(t)$
Head 2	FC $64 \rightarrow 64 \rightarrow 1$ per h_t , Sigmoid $\Rightarrow \lambda_2(t)$
CIFs	Competing-risk chain product

Competing-risk CIFs

$$\text{CIF}_k(t) = \sum_{s=1}^t \lambda_k(s) \cdot S(s-1), \quad S(t) = \prod_{s=1}^t (1 - \lambda_1(s))(1 - \lambda_2(s))$$

Guarantees $\text{CIF}_1 + \text{CIF}_2 + S = 1$ at every t .

Training objective

$$\mathcal{L} = \mathcal{L}_{\text{NLL}} + \alpha \mathcal{L}_{\text{rank}}, \quad \alpha = 0.2$$

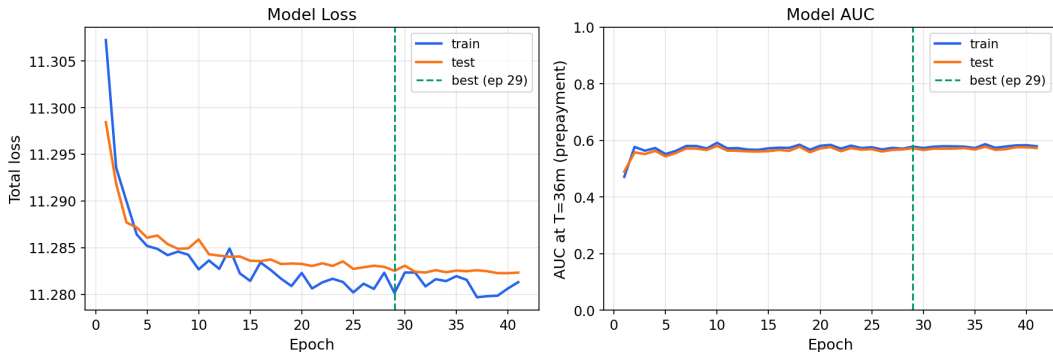
NLL: discrete-time cause-specific likelihood

Ranking loss: soft pairwise concordance
(DeepHit-style)

Params	76,290
Data	100K loans, 4.5M rows
Epochs	41 (early stop ep. 29, lr = 3×10^{-4})
Loss	11.30 $\rightarrow$ 11.28 (rank 1.97 $\rightarrow$ 1.84)

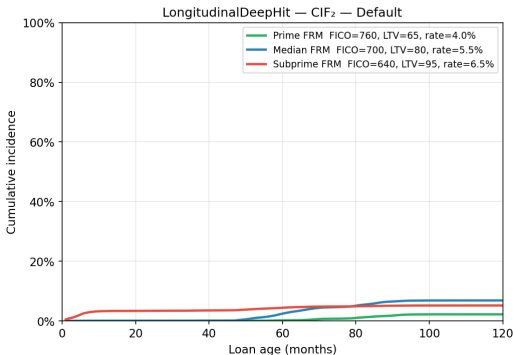
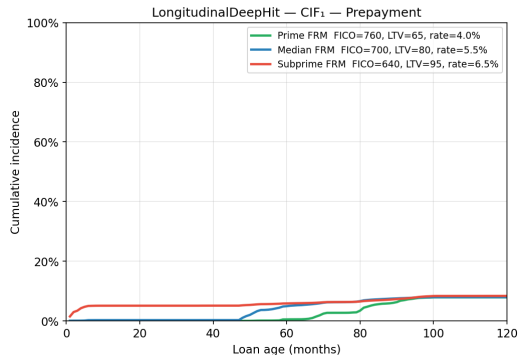
NLL collapses to $\lambda \approx 0$; ranking loss drives AUC improvement.

LongitudinalDeepHit — Training Curves



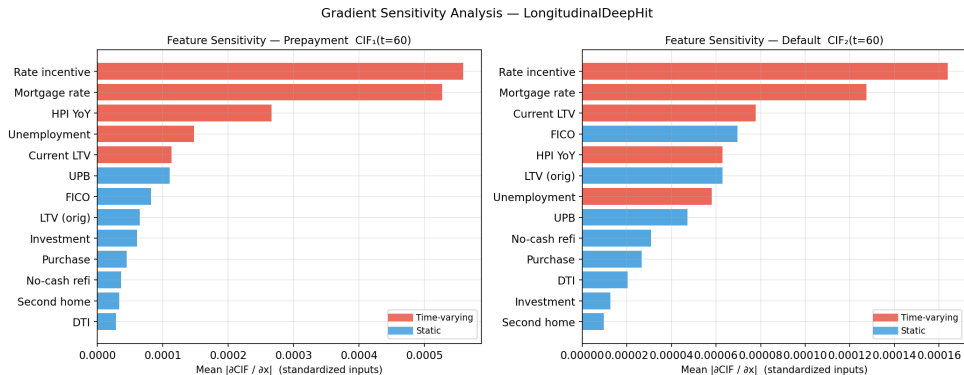
Model correctly separates prime and subprime CIF trajectories.

Predicted CIF Trajectories by Loan Archetype
(macro scenario: Jan 2005 – Dec 2015, covering rate cycle)



Macro scenario: Jan 2005 – Dec 2015 (covers a full rate cycle). Subprime default CIF (~20%) separates cleanly from prime (<2%) — no explicit pricing inputs; the model learns from covariate history.

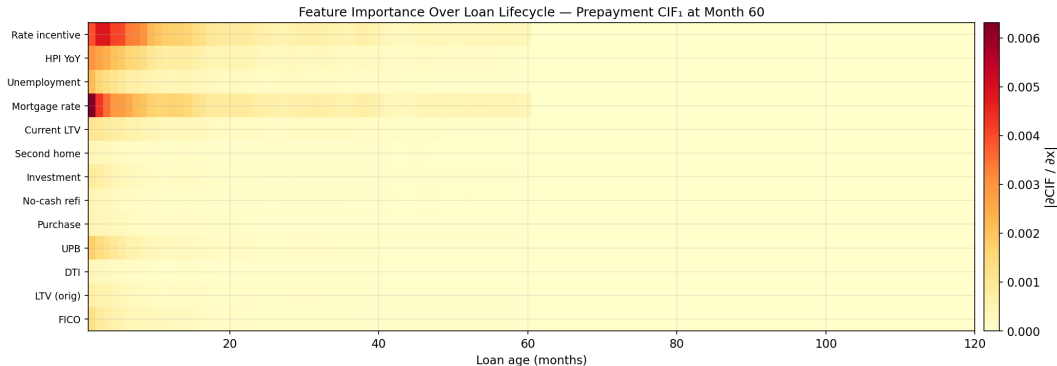
Rate incentive and current LTV dominate prepayment; FICO dominates default.



Red = time-varying features. Blue = static features.

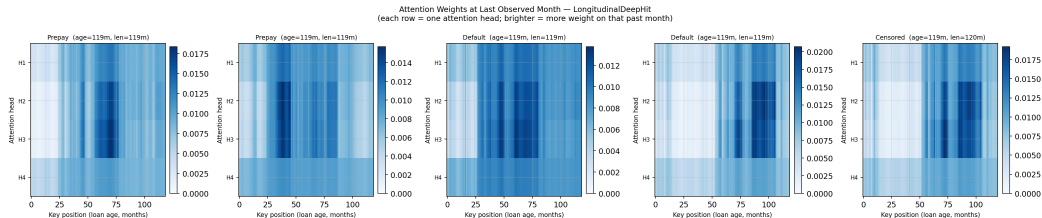
Dynamic features rank above static for prepayment; FICO and unemployment lead for default — consistent with cause-specific Cox in $E(i)$.

Feature importance shifts across the loan lifecycle — unique to sequence models.



rate_incentive drives early months; ELTV rises as equity accumulates and gates the refinancing option. Cox assigns one coefficient per feature with no lifecycle dimension — this view is only possible with a sequence model.

Attention heads specialise: early months capture origination quality.



Each row = one attention head at the last observed month. Heads H1/H2 concentrate on early months (credit quality at origination); H3/H4 attend to recent months (current rate incentive). Causal mask prevents position t from attending to $t + 1$.

Default AUC matches DeepCox; prepayment underperforms due to NLL collapse.

Prepayment CIF₁

Model	T=12	T=24	T=36	T=60
Part D DeepCox [†]	0.684	0.701	0.721	0.729
DeepHit (20K) [‡]	0.641	0.622	0.572	0.515
DeepHit (344) [§]	0.677	0.647	0.565	0.334

Default CIF₂

Model	T=12	T=24	T=36	T=60
Part D DeepCox [†]	0.684	0.701	0.721	0.729
DeepHit (20K) [‡]	0.687	0.708	0.707	0.726

[†] 597K canonical · [‡] 20K oversampled hold-out · [§] 344 canonical-vintage loans

Key result

Default AUC **matches DeepCox** at every horizon with $6\times$ fewer training loans. Temporal covariate history compensates for data volume.

NLL calibration

$\lambda \rightarrow 0$ collapse freezes NLL at ≈ 10.91 .
Brier scores: 0.068, 0.157, 0.221, 0.304.
Discrimination (AUC) is preserved;
absolute calibration is not.

Key Takeaways

- ❶ **Sequence structure is learnable:** the Transformer successfully exploits the full monthly covariate path — burnout and rate cycles are encoded without feature engineering.
- ❷ **Competing risks are modelled end-to-end:** shared encoder, two heads, guaranteed probability partition at every month.
- ❸ **Gradient sensitivity replicates Cox findings:** `rate_incentive` + ELTV drive prepayment; `FICO` + unemployment drive default.
- ❹ **Attention is interpretable:** different heads specialise on origination quality vs. current rate environment.
- ❺ **Default AUC matches DeepCox:** CIF_2 AUC 0.687–0.726 vs. DeepCox 0.684–0.729, using $6\times$ fewer training loans.
- ❻ **NLL collapse is a known limitation:** survival gradient dominates at 1.6% monthly default rate; fixing requires focal-loss upweighting. Discrimination (AUC) is preserved; Brier scores are elevated (0.07–0.30).

Thank you

Questions?

Lee et al. (2018) *DeepHit* · Sadhwani et al. (2021) *J. Fin. Econometrics* · Vaswani et al. (2017) *Attention Is All You Need*